Precision-Retention Frontiers for Low-Precision Analog Vision via Ensemble Hardware-Aware Training

Songqiao Li

Abstract

We present a behavioral simulation framework for analog compute-in-memory inference, focusing on precision-retention trade-offs under realistic physical deployment constraints. The central result is a phase-change-memory (PCM) precision-retention frontier with a non-monotonic precision–accuracy curve: under the tested PCM UnitCell regime, 8-bit PCM achieves the highest fresh accuracy (77.60%) with negligible drift, 4-bit PCM achieves comparable fresh accuracy (76.68%) but violates a 1 pp deployment-drift SLA with a 4.01 pp one-day loss, and 6-bit PCM enters a device-to-device (D2D) sensitive transition zone with lower mean accuracy (68.44%, seed-level std 6.03%) but drift stability (<0.1 pp). To separate this physical frontier from algorithmic robustness, we first study an idealized AIHWKit IdealDevice regime. There, standard fixed-instance hardware-aware training reaches 87.28% fresh-instance accuracy at 8-bit precision but collapses to 14.64% at 4-bit; Ensemble Hardware-Aware Training (HAT), which resamples device-to-device mismatch masks during training, restores the 4-bit pure-quantization ablation to 86.16%. These findings separate the algorithmic challenge of cross-instance robustness from the physical constraint that PCM deployment is gated by precision-dependent retention, with 8-bit emerging as the strongest overall operating point and 6-bit defining a quantization-D2D interaction regime rather than a Pareto optimum.

Keywords: analog compute-in-memory; hardware-aware training; phase-change memory; low-precision inference; precision-retention tradeoff; vision transformers; mixed-signal simulation.

1 Introduction

Compute-in-memory (CIM) architectures execute dense matrix-vector multiplications directly within memory arrays, eliminating the data-movement bottleneck that limits conventional digital systems[1–3]. Across emerging analog memories, including resistive memories, phase-change memory (PCM), and organic optoelectronic devices, deployment accuracy is shaped by finite conductance precision, device-to-device variability, cycle-to-cycle noise, and retention drift[4–8]. These effects are often characterized at the device level, but their system-level consequences for modern vision backbones remain difficult to rank before fabrication.

However, translating device-level models into deployment-facing model accuracy faces a critical methodological gap. Mature CIM simulators expose increasingly detailed analog effects[2;4;5;9], yet the algorithmic and physical failure modes are often entangled: a low-precision model can fail because hardware-aware training overfits one fixed mismatch realization, because quantization is too coarse, or because a realistic memory substrate drifts after programming. Device-specific studies, including organic and optoelectronic demonstrations, further highlight this gap because variability, ADC resolution, retention-induced drift, and transfer across fabricated instances are frequently absent from network-level analyses or treated only qualitatively[10–12]. A deployment workflow therefore needs to separate algorithmic robustness from physical precision-retention limits.

Here we present a behavioral simulation framework that evaluates analog neural network deployment under both idealized pure-quantization and realistic device-physics regimes. We apply a mixed-signal deployment model to the Tiny-ViT-5M backbone to systematically disentangle algorithmic generalization failures from physical retention limits. Three concrete contributions follow. First, we map a 4/6/8-bit precision-retention deployment frontier for PCM and uncover a non-monotonic precision–accuracy curve. In this tested sweep, 4-bit PCM is trainable but drift-limited, 8-bit PCM is drift-flat with the highest fresh accuracy (77.60%), and 6-bit PCM enters a D2D-sensitive transition zone with lower mean accuracy (68.44%, seed-level std 6.03%) but drift stability, reflecting a quantization-D2D interaction where device mismatch perturbations become comparable to quantization steps at 1/64 resolution. Second, we expose why the frontier cannot be inferred from algorithm-only tests: in an idealized pure-quantization regime, standard fixed-instance hardware-aware training at 4-bit collapses to near-chance, while Ensemble Hardware-Aware Training (HAT) restores fresh-instance accuracy to 86.16±0.19% by resampling device-to-device mismatch masks each epoch. Third, we use the contrast between the ideal-device ablation and the PCM ladder to show that algorithmic cross-instance robustness and physical precision-retention selection are separate deployment decisions.

The framework is intentionally behavioral rather than circuit-accurate, focusing on macroscopic precision and drift phenomena rather than array-level parasitics (IR drop, sneak paths) which remain crucial future considerations. Additional supporting analyses, including front-end compensation and zero-shot transfer to alternative organic optoelectronic profiles, are provided in the Supplementary Information. Its purpose is to provide a reproducible workflow for comparing mitigation strategies and establishing physical deployment frontiers.

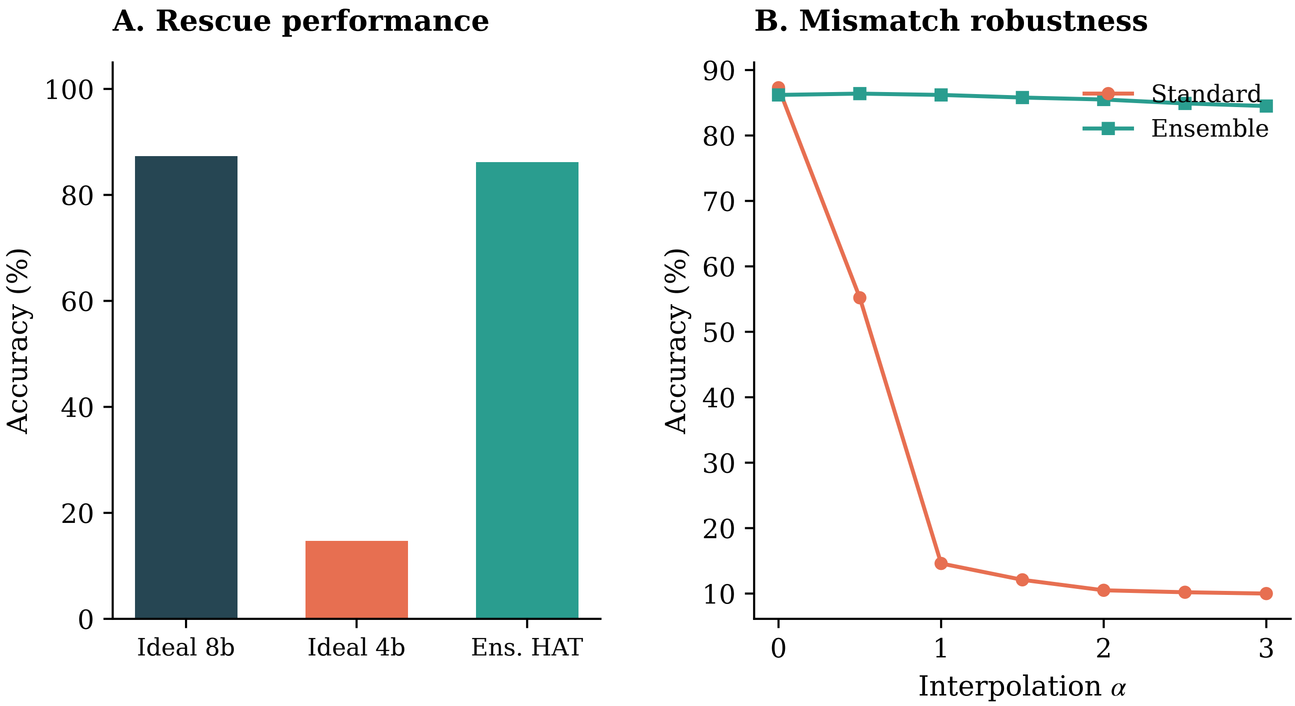

Figure 1: Algorithmic generalization failure in the ideal-device ablation. (A) Fresh-instance accuracy for the AIHWKit IdealDevice baseline. 4-bit pure quantization completely collapses, but Ensemble HAT restores performance near the 8-bit baseline. This is an algorithmic ablation, not the PCM deployment headline. (B) Loss-landscape interpolation from the source training mask ($\alpha = 0$) to fresh hardware instances ($\alpha = 1$). Ensemble HAT discovers a broad plateau robust to D2D mismatch, while Standard HAT exhibits extreme sensitivity.

2 Related Work

2.1 Hardware-Aware Training and Robustness

Hardware-aware training (HAT) mitigates analog non-idealities by injecting hardware perturbations into the forward path during optimization[13]. Although related in spirit to quantization-aware training[14] and to domain randomization in sim-to-real transfer[15], HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for a given hardware instance. It is therefore different from independent thermal or sampling noise. Ensemble HAT resamples the static D2D mask at each training epoch, improving zero-shot transfer to fresh hardware instances. The connection to domain randomization is deliberate, but the analogy is incomplete because hardware-induced perturbations decompose into D2D mismatch—spatially structured and fixed for a given fabricated instance—and cycle-to-cycle (C2C) noise—stochastic per-read with zero mean.

2.2 Organic Neuromorphic and Optoelectronic Devices

Organic synaptic and memristive devices offer optical sensitivity, multilevel conductance tuning, and flexible-substrate compatibility for neuromorphic hardware[6;8;16;17]. Recent demonstrations include multilevel memory[7;11], long retention tails[10], and integrated optoelectronic arrays[18–21]. Array-level studies are beginning to treat active-matrix addressing, spatial optical response, and crosstalk as system constraints rather than purely materials observations[22–24], while temperature-resilient organic synapses and high-temperature synaptic phototransistors suggest that thermal stability is also becoming deployment-relevant[25;26].

2.3 CIM Simulation Frameworks and Hybrid Mapping

System-level CIM simulators such as DNN+NeuroSim established an effective template for connecting device assumptions, peripheral-circuit costs, and neural-network accuracy[2]. MemTorch showed how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation[9], AIHWKIT exposed analog HAT and inference simulation in a practical mixed-signal software stack[4], and CrossSim provides a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models[5]. These mature simulators were developed primarily around inorganic resistive memories, and they do not directly expose all profile-level effects of interest, such as inverse-gamma optoelectronic photoresponse or organic-specific double-exponential retention, as first-class device primitives. Our behavioral profile-driven approach therefore serves as a profile-level extension and deployment-risk ranking layer rather than as a replacement for established CIM toolkits.

3 Methodology

3.1 Hybrid Analog/Digital Inference Stack

We adopt a hybrid analog/digital inference stack: dense linear operators execute on differential-pair crossbars; control-heavy operators remain digital. Concretely, the analog partition contains patch embeddings, query/key/value (Q/K/V) and output projections, and feed-forward layers; the digital partition contains softmax, layer normalization, GELU activation, positional encoding, residual additions, and the class token, following established practice in heterogeneous CIM accelerators[27–31].

Under this mapping, approximately 88% of parameters execute on analog arrays while roughly 60% of estimated energy remains digital, largely because attention operations stay in the digital domain—a pattern consistent with recent heterogeneous CIM accelerators for Vision Transformers[28–31].

Table 1: Canonical Simulation Parameters and Hardware Constraints. These values define the default non-ideality regime for the Tiny-ViT-5M experiments unless otherwise specified.

Parameter	Value	Physical Meaning
n_{states}	16	Conductance resolution (4-bit equivalent)
σ_{D2D}	10%	Device-to-device mismatch (relative std. dev.)
σ_{C2C}	5%	Cycle-to-cycle read noise
$G_{\max}/G_{\min}$	10	Dynamic range of analog weights
ADC bits	8-bit	Default readout quantization (before sweep)

3.2 Modeling Physical Non-Idealities

The framework injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight is decomposed into positive and negative branches, mapped to the conductance window $[G_{\min}, G_{\max}]$, and quantized to n_{states} levels via a straight-through estimator (STE) quantizer $Q_n(\cdot)$. Differential readout recovers $G_{\text{eff}} = \tilde{G}^{+} - \tilde{G}^{-}$, and the downstream digital weight is

$$\hat{W}_{ij} = s_\ell G_{\text{eff},ij}, \qquad s_\ell = \begin{cases} \dfrac{w_{\max}}{G_{\max} - G_{\min}}, & \text{standard calibration} \\ \dfrac{w_{\max}}{\max_{i,j} |G_{\text{eff},ij}| + \varepsilon}, & \text{retention-time recalibration} \end{cases} \tag{1}$$

with layer index ℓ omitted. Unless otherwise stated, we use the standard-calibration branch; the retention-recalibration branch is used only for specific retention-drift diagnostics. The recovered weight step is $\Delta_{\hat{W}} = s_\ell (G_{\max} - G_{\min})/(n_{\text{states}} - 1)$, so perturbations smaller than half a step are masked. The differential-pair scheme suppresses common-mode noise[32;33].

Hardware-aware training (HAT) injects forward perturbations while back-propagating through a straight-through estimator. Standard HAT fixes one D2D mismatch field M_0 during training.

Ensemble HAT as distribution matching. Ensemble HAT treats the mismatch map as a random variable resampled each epoch:

$$\theta^{\star}_{\text{ens}} = \arg\min_{\theta} \mathbb{E}_{M \sim \mathscr{D}_{\text{D2D}}} \left[\frac{1}{N} \sum_{n=1}^{N} \ell\big(f(x_n; \theta, M), y_n\big) \right], \tag{2}$$

where the expectation is Monte-Carlo estimated with one fresh draw $M^{(e)} \sim \mathscr{D}_{\text{D2D}}$ held constant across all mini-batches in epoch e. Expanding Eq. 2 (Supplementary Note S-Theory) reveals an implicit Fisher-weighted gradient-squared regularization effect that pushes the optimizer toward flat regions in the M-direction, explaining the fresh-instance transferability without extra hyperparameters. Epoch-level (not per-batch) resampling is load-bearing, giving the optimizer enough consistent signal to fit each instance before resampling[15]. Fresh-instance transfer is then evaluated on unseen i.i.d. D2D draws held constant for each evaluation pass. For a single checkpoint, the reported fresh-instance spread is the standard deviation across ten fresh-instance means (each averaged over five forward passes); for the primary results, the uncertainty is the sample standard deviation across three seed-level fresh-instance means[1].

[1]The canonical single-seed headline used in the thesis is 86.37 ± 1.54%; this study reports the 3-seed mean 86.16 ± 0.19% to emphasize reproducibility robustness.

Algorithm 1 Ensemble Hardware-Aware Training (HAT)

procedure Train($\mathscr{D}_{\text{train}}, \eta, \mathscr{D}_{\text{D2D}}$)
  Initialize weights $\boldsymbol{\theta}$
  for epoch $e = 1 \ldots E$ **do**
    Sample fresh mismatch mask $M^{(e)} \sim \mathscr{D}_{\text{D2D}}$
    for batch $(x, y) \in \mathscr{D}_{\text{train}}$ **do**
      Compute output with current mask: $\hat{y} = f(x; \boldsymbol{\theta}, M^{(e)})$
      Update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \ell(\hat{y}, y)$
    end for
  end for
  return $\boldsymbol{\theta}$
end procedure
procedure Evaluate($x, \boldsymbol{\theta}, \mathscr{D}_{\text{D2D}}$)
  Sample unseen masks $\{M'_k\}_{k=1}^{K} \sim \mathscr{D}_{\text{D2D}}$
  return $\frac{1}{K} \sum_{k=1}^{K} \text{Acc}(f(x; \boldsymbol{\theta}, M'_k))$
end procedure

3.3 Training Protocol

All Tiny-ViT experiments use AdamW ($lr = 5 \times 10^{-4}$, weight decay 0.05), 100 epochs, batch size 64, cosine annealing ($T_{\max} = 100$). Images are resized to 224×224 with standard augmentation. The default analog resolution is $n_{\text{states}} = 16$ levels. We evaluate several model configurations: a digital reference, a hybrid baseline without noise, a fixed-mismatch noisy baseline, and the proposed Ensemble HAT regime. Additional variants incorporate front-end models and retention-drift corrections for specific diagnostic checks (see Supplementary Note S-Notation).

Nonlinear-write and retention surrogates are anchored to measured DNTT transients[34]. Nonlinear write modifies the STE backward pass by scaling the straight-through gradient with a state-dependent factor proportional to the present conductance magnitude raised to $NL - 1$[2]; ideal write is recovered when $NL_{\text{LTP}} = 1$ (long-term potentiation) and $|NL_{\text{LTD}}| = 1$ (long-term depression). The optional optoelectronic front-end model applies inverse-gamma preprocessing, photocurrent shot-noise injection, and per-sample renormalization; the explicit response equations are kept in the Supplementary Information.

3.4 Profile-Driven Substitution Interface

The profile interface packages technology-specific assumptions into a replaceable JSON parameter bundle. Deployment holds the digital backbone and training recipe fixed while substituting the profile, enabling literature priors today and measured-device calibration later without code changes.

3.5 Sensitivity and Energy Metrics

For the joint D2D–ADC sweep, first-order factor importance is quantified with Sobol indices estimated from the 7×9 grid of Monte Carlo means. Energy is estimated via a first-order analytical model (Supplementary Methods) from operator counts and literature-proxy constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ); these are analytical proxy values for relative comparison, not measured silicon values.

[2] The absence of an explicit NL prefactor is intentional: this is a behavioral proxy for position-dependent update difficulty, not the strict derivative of a power-law $f(G) = G^{NL}$.

4 Experimental Setup

We evaluate ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M backbones across three classification benchmarks: CIFAR-10, CIFAR-100, and Flowers-102. These models establish a progression from entry-level validation to high-capacity transformer deployment. The digital FP32 baselines for these configurations are summarized in Table 2. Models for CIFAR-10/100 were trained from scratch, while Flowers-102 utilized ImageNet-pretrained weights.

Table 2: Evaluated backbones and digital (FP32) accuracy baselines. Data establishes the ideal software performance before analog non-ideality injection.

Dataset	ResNet-18	ConvNeXt-Tiny	Tiny-ViT-5M
CIFAR-10	95.46%	90.74%	98.06%
CIFAR-100	78.64%	64.12%	86.94%
Flowers-102	n.a.	33.22%	97.97%

The main text evaluation focuses on the Tiny-ViT-5M backbone under extreme low precision (4-bit) and the realistic phase-change memory (PCM) precision ladder (4/6/8-bit). Detailed optimization settings, Monte Carlo evaluation protocols, and auxiliary architectural experiments (such as ConvNeXt retention and proportional-noise studies) are summarized in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information, together with the optimization settings and evaluation conditions used for the reported runs. The present study therefore targets reproducible reruns through archived configurations, logs, and model lineage. We also emphasize that the framework operates at the simulation level: all device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. Validation against physical hardware remains for future work, and the quantitative results below should be interpreted as comparative risk rankings rather than absolute deployment predictions.

5 Results

The results are organized around two questions: whether training can generalize across unseen hardware instances, and which physical precision regime remains deployable once that algorithmic failure mode is controlled.

5.1 IdealDevice Ablation: Fixed-Mask Failure and Algorithmic Rescue

We first measure the behavior of the tested AIHWKit IdealDevice quantization/noise baseline as an algorithmic ablation, not as the physical PCM deployment result. The fixed-mask HAT protocol mirrors the common hardware-aware-training practice of optimizing against one sampled hardware instance or analog tile configuration[4;13]; here we replicate that protocol and test its failure boundary. As summarized in Table 3, standard per-batch hardware-aware training achieves 87.28% fresh-instance accuracy at 8-bit precision, but completely collapses to 14.64% at 4-bit (Figure 1A). This indicates a severe failure to generalize under fresh hardware instance shifts at extreme low precision.

To isolate the sources of error, we evaluate the system under sequential constraint removal (Table 4). Quantization and hybrid mapping alone (without noise) introduce only a minimal penalty (98.06% $\rightarrow$ 97.39%). The primary failure is triggered by device-to-device mismatch, which Ensemble HAT successfully mitigates.

By resampling the device-to-device (D2D) variation patterns during the forward pass (Figure 1A), Ensemble Hardware-Aware Training (HAT) rescues this 4-bit pure-quantization ablation. The model recovers to 86.16% fresh-instance accuracy, effectively closing the gap with

Table 3: IdealDevice pure-quantization ablation. Data shows fresh-instance accuracy across three seeds for the AIHWKit IdealDevice model. These results establish algorithmic cross-instance robustness, not PCM physical deployment performance.

Model Configuration	Precision	Fresh Accuracy	Role
IdealDevice Baseline	8-bit	$87.28 \pm 0.13\%$	Stable upper bound
IdealDevice Baseline	4-bit	$14.64 \pm 0.11\%$	Generalization collapse
Ensemble HAT	4-bit	$86.16 \pm 0.19\%$	Algorithmic rescue

Table 4: Ablation of mapping and noise constraints. Accuracy reported for the Tiny-ViT-5M backbone at 4-bit precision (where applicable) on CIFAR-10.

Configuration	Precision	Partition	Mismatch Mask	Accuracy
Digital Reference	32-bit	Digital	None	$98.06 \pm 0.14\%$
Hybrid Baseline	4-bit	Analog/Digital	None	$97.39 \pm 0.08\%$
Fixed-mask HAT	4-bit	Analog/Digital	Single (fixed)	$10.00 \pm 0.00\%$
Ensemble HAT	4-bit	Analog/Digital	Resampled	$86.16 \pm 0.19\%$

the 8-bit ideal-device baseline and demonstrating robust cross-instance transfer. This result establishes that 4-bit operation is algorithmically viable when device physics are idealized. It does not imply that 4-bit PCM is drift-stable. Loss-landscape interpolation along the D2D mismatch direction (Figure 1B) confirms that Ensemble HAT discovers a robust plateau, whereas Standard HAT falls into a sharp, instance-specific local minimum.

5.2 Realistic PCM Training and the Precision-Retention Frontier

While Ensemble HAT addresses the algorithmic generalization problem in a pure-quantization context, realistic analog deployment must account for complex device physics, particularly conductance drift. We evaluate our approach using the `PCMPresetUnitCell` model in AIHWKit, which incorporates non-linear update physics, finite dynamic range, and time-dependent drift.

Under the tested PCM UnitCell simulation regime, training is viable across 4-bit, 6-bit, and 8-bit precisions. However, the results reveal a non-monotonic precision–accuracy curve and a precision-retention deployment frontier. We mark a setting as drift-constrained when its 24-hour accuracy loss exceeds a 1 pp deployment SLA; by this criterion, 8-bit and 6-bit PCM remain effectively drift-flat, whereas the 4-bit configuration incurs a 4.01 pp retention penalty and is therefore drift-limited (Table 5). Notably, 6-bit PCM achieves the lowest mean fresh accuracy despite its drift stability, reflecting a quantization-D2D interaction: at 1/64 resolution, device mismatch perturbations become comparable to quantization steps, allowing the model to overfit to its training-time D2D mask. At 4-bit, quantization noise masks D2D variability; at 8-bit, the fine grid absorbs it.

Table 5: PCM Precision Ladder and Stability Frontier. Fresh accuracy reflects corrected new-protocol evaluations (3 seeds for 4/8-bit; 4 seeds for 6-bit). The retention columns are from the drift-evaluation protocol; Δ drift is computed as retention-eval 0 s accuracy minus 24 h accuracy, not as fresh-eval mean minus 24 h accuracy. The 6-bit row uses the completed four-seed drift closure and shows drift stability but low mean fresh accuracy, consistent with a D2D-sensitive transition regime rather than a deployment frontier optimum.

Precision	Fresh Acc.	1 h Acc.	24 h Acc.	$\Delta_{0\to 24h}$	Deployment Role
8-bit PCM	77.60%	77.49%	77.57%	0.04 pp	Drift-flat reference
6-bit PCM	68.44%	68.46%	68.36%	0.04 pp	D2D-sensitive transition zone
4-bit PCM	76.68%	74.04%	72.64%	4.01 pp	Drift-limited regime

5.3 Iso-Accuracy Operating Envelope

The architectural operating envelope becomes clear when ADC precision and device-to-device variability are swept jointly (Figure 2). Accuracy is near chance below 5-bit ADC, and the transition from 5 to 6 bits produces a consistent ∼7 pp gain. Once readout precision reaches this threshold, further degradation is governed mainly by D2D variability rather than by ADC quantization. The Sobol analysis captures the same hierarchy: $S_{ADC} = 0.98$ over the full grid, whereas $S_{D2D} = 0.92$ after restricting to the operational regime. The practical implication is therefore sequential rather than simultaneous: first secure 6-bit readout, then optimize device variability.

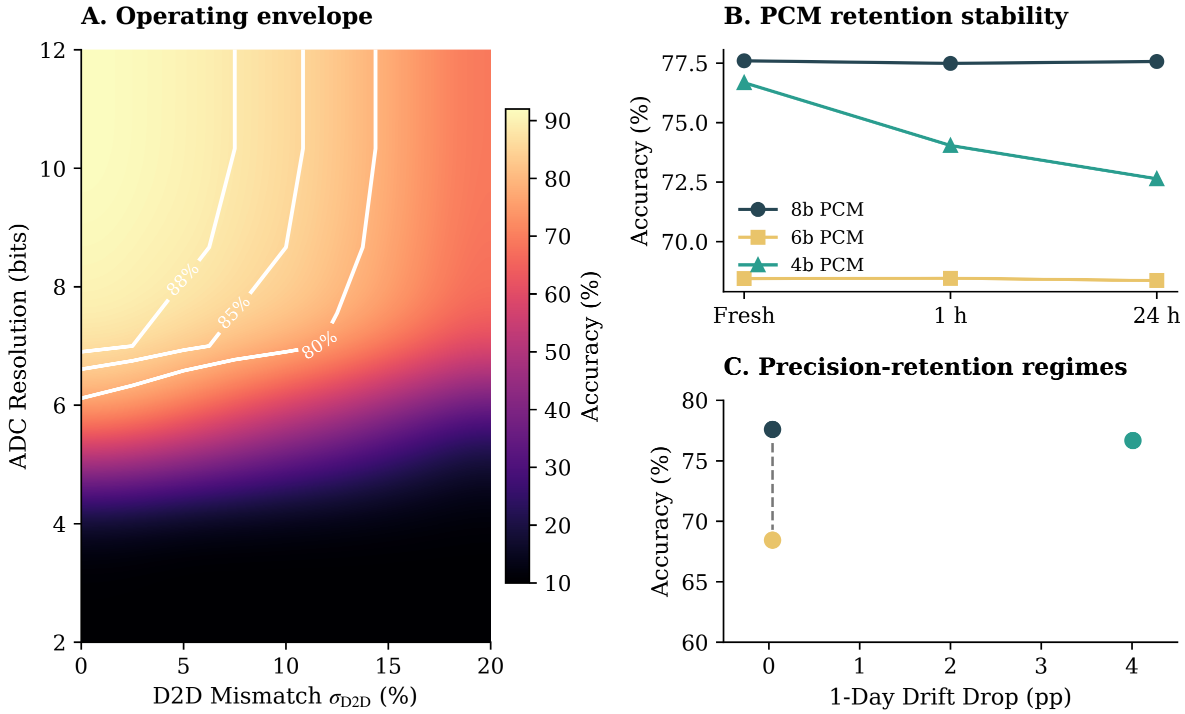

Figure 2: Physical constraints and the iso-accuracy operating envelope. Contour map under joint ADC precision and D2D mismatch sweep. Readout precision forms a hard cliff below 6 bits, above which D2D variability dictates the operational boundary. Bold contour lines indicate the 80%, 85%, and 88% accuracy levels.

5.4 Deployment Guidance

The precision ladder separates the algorithmic challenge of cross-instance robustness from the device-level precision-retention tradeoff, revealing a non-monotonic relationship between weight precision and fresh accuracy. 8-bit PCM is the strongest operating point: 77.60% fresh accuracy with negligible drift. 4-bit PCM achieves comparable fresh accuracy (76.68%) but loses ∼4.01 pp over one day and violates the 1 pp drift SLA. 6-bit PCM, while drift-stable, exhibits the lowest mean fresh accuracy (68.44%) and high seed-to-seed variance (std 6.03%), entering a D2D-sensitive transition zone where device mismatch perturbations are comparable to quantization steps at 1/64 resolution.

5.5 Future Directions

Extending these hardware-aware training principles to memory-bound inference components, such as analog KV-cache, presents a compelling future trajectory for large-scale deployment.

6 Discussion

These findings should be read as a separation of deployment decisions: training must first avoid hardware-instance overfitting, after which physical precision, drift, and readout limits determine the usable operating point.

6.1 Separating Algorithmic and Physical Frontiers

The primary finding of this behavioral simulation study is the clear separation between algorithmic generalization risks and physical device limits in analog CIM deployment. Standard fixed-mask hardware-aware training, while effective at higher precisions, fails fundamentally at extreme low precision (4-bit) because the network overfits to a single hardware instance's mismatch field. Ensemble HAT resolves this by treating hardware variability as a distribution-level regularizer during training, effectively "rescuing" the 4-bit regime for hardware instances unseen during training.

This rescue, however, does not eliminate physical constraints; it merely reveals them. Once algorithmic convergence is secured, the system is bounded by the precision-retention frontier of the underlying memory technology. In our tested PCM UnitCell models, 8-bit precision emerges as the strongest practical operating point, combining high fresh accuracy (77.60%) with drift stability. 6-bit precision, while drift-stable, enters a D2D-sensitive regime where device mismatch degrades mean accuracy (68.44%) and introduces high seed-to-seed variance (std 6.03%)—a transition zone between the quantization-dominated 4-bit regime and the noise-absorbed 8-bit regime.

6.2 Hard Cliffs and Operational Envelopes

The joint ADC–D2D sensitivity analysis identifies a hierarchical set of hardware requirements. ADC resolution serves as a "hard gate": below 6 bits, classification accuracy collapses regardless of device quality. Above this threshold, device-to-device variability becomes the primary determinant of accuracy. This hierarchy suggests a clear hardware roadmap for vision transformer deployment: prioritize a high-precision (6-bit or greater) readout interface before attempting to minimize device-level mismatch.

6.3 Limitations and outlook

Scope. The conclusions are bounded by the present behavioral simulation stack. Device profiles are calibrated from literature values or a single fabrication batch, so the numbers do not claim cross-batch, temperature, aging, or circuit-layout coverage. We model quantization, D2D/C2C variability, ADC precision, write nonlinearity, and retention, but defer parasitic IR drop, sneak paths, and pulse-faithful programming to follow-on hardware-calibrated studies.

Energy and scale. The energy estimates are first-order analytical projections from literature-proxy constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ; Section 3.5). They rank deployment risks rather than predict silicon performance. The current ablation matrix also prioritizes non-ideality coverage over dataset breadth; cross-architecture TinyImageNet sweeps (see Supplementary Information) are treated as validation, not as main-text claims.

Outlook. The next tests are straightforward: replace UnitCell presets with measured-array statistics, verify whether the non-monotonic precision-accuracy curve and 6-bit transition-zone behavior persist under measured-device statistics, and evaluate memory-bound inference components such as analog KV-cache with explicit train/eval noise separation. These extensions do

Table 6: Deployment Risk Ranking and Mitigation Strategies. Factors are ranked by their impact on categorical accuracy as identified by Sobol sensitivity analysis and precision-ladder sweeps.

Factor	Impact Rank	Mitigation Strategy
ADC Resolution	High (Hard Gate)	Secure $\geq$6-bit readout precision
Stochastic D2D	High (Generalization)	Ensemble HAT (distribution matching)
PCM Drift	Moderate (Time-bound)	8-bit for stability; refresh 4-bit periodically
Write Nonlinearity	Moderate (Optimization)	STE gradient scaling
Shot Noise	Low (Transduction)	Front-end inverse-gamma compensation

not alter the present conclusion: algorithmic cross-instance resampling and physical precision-retention selection are separate deployment decisions.

7 Conclusion

We presented a behavioral simulation framework for analog compute-in-memory inference that disentangles algorithmic generalization failures from physical precision-retention constraints. By comparing ideal pure-quantization baselines against realistic phase-change memory (PCM) physics, the framework serves as a practical decision aid for establishing deployment frontiers in low-precision analog hardware.

Several observations stand out. Under the tested PCM UnitCell simulation regime, convergence is observed across the 4/6/8-bit precision ladder where pure quantization fails. This convergence exposes a non-monotonic precision–accuracy curve: 8-bit PCM achieves the highest fresh accuracy (77.60%) with negligible drift, 4-bit PCM achieves comparable fresh accuracy (76.68%) at the cost of a 4.01 pp one-day drift penalty, and 6-bit PCM enters a D2D-sensitive transition zone with lower mean accuracy (68.44%, seed-level std 6.03%) but drift stability. The 6-bit dip reflects a quantization-D2D interaction where, at 1/64 resolution, device mismatch perturbations become comparable to quantization steps. Separately, in the ideal-device ablation, fixed hardware-instance transfer is the most prominent algorithmic risk observed in our matrix: naive 4-bit deployment collapses to 14.64%, whereas Ensemble Hardware-Aware Training (HAT) resamples mismatch masks to recover fresh-instance accuracy to 86.16$\pm$0.19%.

Ultimately, this framework clarifies that standard fixed-mask training is insufficient for extreme low-precision analog deployment. Ensemble HAT resolves this algorithmic bottleneck, allowing system designers to navigate the subsequent physical limits of the analog precision-retention frontier. Our findings suggest that noise-aware training principles are foundational for memory-bound inference components, including analog KV-cache architectures.

References

[1] Mark Horowitz. Computing's energy problem (and what we can do about it). In IEEE International Solid-State Circuits Conference (ISSCC), pages 10–14, 2014. doi: 10.1109/ISSCC.2014.6757323. URL https://doi.org/10.1109/ISSCC.2014.6757323.

[2] Xiaochen Peng, Shanshi Huang, Hongwu Jiang, Anni Lu, and Shimeng Yu. DNN+NeuroSim v2.0: An end-to-end benchmarking framework for compute-in-memory accelerators for on-chip training. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 40(11):2306–2319, 2021. doi: 10.1109/TCAD.2020.3043731. URL https://doi.org/10.1109/TCAD.2020.3043731.

[3] Mario Lanza et al. The growing memristor industry. Nature, 640:613–622, 2025. doi: 10.1038/s41586-025-08733-5.

[4] Malte J. Rasch, Diego Moreda, Tayfun Gokmen, Manuel Le Gallo, Fabio Carta, Cindy Goldberg, Kaoutar El Maghraoui, Abu Sebastian, and Vijay Narayanan. A flexible and fast PyTorch toolkit for simulating training and inference on analog crossbar arrays, 2021. URL https://arxiv.org/abs/2104.02184.

[5] Sandia National Laboratories. Crosssim: Sandia's simulator for analog ai accelerators. Technical report, Sandia National Laboratories, 2024. URL https://doi.org/10.2172/2585829.

[6] Yunchao Xu, Yuan He, Dongyong Shan, Biao Zeng, and Qian-Xi Ni. Emerging artificial synaptic devices based on organic semiconductors: Molecular design, structure and applications. ACS Applied Materials & Interfaces, 2025. doi: 10.1021/acsami.4c17455. URL https://doi.org/10.1021/acsami.4c17455.

[7] Haotian Guo, Jing Guo, Yujing Wang, Hezhen Wang, Simin Cheng, Zehao Wang, Qian Miao, and Xiaomin Xu. An organic optoelectronic synapse with multilevel memory enabled by gate modulation. ACS Applied Materials & Interfaces, 2024. doi: 10.1021/acsami.3c19624. URL https://doi.org/10.1021/acsami.3c19624.

[8] Li Zhang, Wei Chen, and Jing Xu. Recent progress in organic optoelectronic synaptic devices. Photonics, 12(5):435, 2025. doi: 10.3390/photonics12050435. URL https://doi.org/10.3390/photonics12050435.

[9] Corey Lammie, Wei Xiang, Bernabé Linares-Barranco, and Mostafa Rahimi Azghadi. Memtorch: An open-source simulation framework for memristive deep learning systems. Neurocomputing, 485:124–133, 2022. doi: 10.1016/j.neucom.2022.02.043. URL https://doi.org/10.1016/j.neucom.2022.02.043.

[10] Jianmin Zeng, Xinhui Chen, Shuzhi Liu, Qilai Chen, and Gang Liu. Organic memristor with synaptic plasticity for neuromorphic computing applications. Nanomaterials, 13(5): 803, 2023. doi: 10.3390/nano13050803. URL https://doi.org/10.3390/nano13050803.

[11] Uihoon Jung, Miseong Kim, Jaewon Jang, Jin-Hyuk Bae, In Man Kang, and Sin-Hyung Lee. Formation of cluster-structured metallic filaments in organic memristors for wearable neuromorphic systems with bio-mimetic synaptic weight distributions. Advanced Science, 11(9):2307494, 2024. doi: 10.1002/advs.202307494. URL https://doi.org/10.1002/advs.202307494.

[12] Yu-Pu Lin, Christopher H. Bennett, Théo Cabaret, Damir Vodenicarevic, Djaafar Chabi, Damien Querlioz, Bruno Jousselme, Vincent Derycke, and Jacques-Olivier Klein. Physical realization of a supervised learning system built with organic memristive synapses. Scientific Reports, 2016. doi: 10.1038/srep31932. URL https://doi.org/10.1038/srep31932.

[13] Vinay Joshi, Manuel Le Gallo, Simon Haefeli, Irem Boybat, S. R. Nandakumar, Christophe Piveteau, Martino Dazzi, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Accurate deep neural network inference using computational phase-change memory. Nature Communications, 11:2473, 2020. doi: 10.1038/s41467-020-16108-9. URL https://doi.org/10.1038/s41467-020-16108-9.

[14] Jungwook Choi, Zhuo Wang, Swagath Venkataramani, Pierce I-Jen Chuang, Vijayalakshmi Srinivasan, and Kailash Gopalakrishnan. PACT: Parameterized clipping activation for quantized neural networks. In Proceedings of the 35th International Conference on Machine Learning, volume 80 of Proceedings of Machine Learning Research, pages 1028–1037, 2018. doi: 10.48550/arXiv.1805.06085. URL http://proceedings.mlr.press/v80/choi18a.html.

[15] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pages 23–30, 2017. doi: 10.1109/IROS.2017.8202133. URL https://doi.org/10.1109/IROS.2017.8202133.

[16] Pietro Belleri, Judith Pons i Tarrès, Iain McCulloch, Paul W. M. Blom, Zsolt M. Kovács-Vajna, Paschalis Gkoupidenis, and Fabrizio Torricelli. Unravelling the operation of organic artificial neurons for neuromorphic bioelectronics. Nature Communications, 15:5350, 2024. doi: 10.1038/s41467-024-49668-1. URL https://doi.org/10.1038/s41467-024-49668-1.

[17] Junpeng Ji, Dace Gao, Han-Yan Wu, Miao Xiong, Nevena Stajkovic, Claudia Latte Bovio, Chi-Yuan Yang, Francesca Santoro, Deyu Tu, and Simone Fabiano. Single-transistor organic electrochemical neurons. Nature Communications, 16:4334, 2025. doi: 10.1038/s41467-025-59587-4. URL https://doi.org/10.1038/s41467-025-59587-4.

[18] Lan Shen Hu, Marco Fattori, Winston Schilp, Roy Verbeek, Setareh Kazemzadeh, Yoeri van de Burgt, Auke Jisk Kronemeijer, Gerwin Gelinck, and Eugenio Cantatore. An energy-efficient solid-state organic device array for neuromorphic computing. IEEE Transactions on Electron Devices, 70(12):6520–6525, 2023. doi: 10.1109/TED.2023.3327947. URL https://doi.org/10.1109/TED.2023.3327947.

[19] Xu Liu, Shilei Dai, Yiyang Jin, Junyao Zhang, Ziyi Guo, Tongrui Sun, Li Li, Pu Guo, Huaiyu Gao, Haixia Liang, Shiqi Zhang, Lize Xiong, Yanmin Zhou, and Jia Huang. Near-infrared organic photoelectrochemical synaptic transistors by wafer-scale photolithography for neuromorphic visual system. Nature Communications, 17:197, 2026. doi: 10.1038/s41467-025-66891-6. URL https://doi.org/10.1038/s41467-025-66891-6. Published online 2025-12-08.

[20] Fanqing Zhang, Xinyi Shan, Zhenhua Ji, Yan Wang, Jing Zhao, and Zhongrui Wang. Large-scale high uniform optoelectronic synapses array for artificial visual neural network. Microsystems & Nanoengineering, 11:42, 2025. doi: 10.1038/s41378-024-00859-2. URL https://doi.org/10.1038/s41378-024-00859-2.

[21] Heyi Huang, Xiangpeng Liang, Yuyan Wang, Jianshi Tang, Yuankun Li, Yiwei Du, Wen Sun, Jianing Zhang, Peng Yao, Xing Mou, Feng Xu, Jinzhi Zhang, Yuyao Lu, Zhengwu Liu, Jianlin Wang, Zhixing Jiang, Ruofei Hu, Ze Wang, Qingtian Zhang, Bin Gao, Xuedong Bai, Lu Fang, Qionghai Dai, Huaxiang Yin, He Qian, and Huaqiang Wu. Fully integrated multi-mode optoelectronic memristor array for diversified in-sensor computing. Nature Nanotechnology, 20(1):93–103, 2025. doi: 10.1038/s41565-024-01794-z. URL https://doi.org/10.1038/s41565-024-01794-z.

[22] Ying Liu, Biao Wang, Lei Wu, Lulu Huang, Lu Lin, Li Xiang, Dongqing Liu, Shiguo Zhang, Chenguang Zhu, and Yijie Tao. Artificial visual synaptic architecture with high-linearity light-modulated weight for optoelectronic neuromorphic computing. ACS Applied Materials & Interfaces, 2023. doi: 10.1021/acsami.3c11495. URL https://doi.org/10.1021/acsami.3c11495.

[23] Dingwei Li, Yitong Chen, Huihui Ren, Yingjie Tang, Siyu Zhang, Yan Wang, Lixiang Xing, Qi Huang, Lei Meng, and Bowen Zhu. An active-matrix synaptic phototransistor array for in-sensor spectral processing. Advanced Science, 11, 2024. doi: 10.1002/advs.202406401. URL https://doi.org/10.1002/advs.202406401.

[24] Tianqing Wan, Bangjie Shao, Sijie Ma, Yue Zhou, Qiao Li, and Yang Chai. In-sensor computing: Materials, devices, and integration technologies. Advanced Materials, 35(37): 2203830, 2023. doi: 10.1002/adma.202203830. URL https://doi.org/10.1002/adma.202203830.

[25] A. Melianas, T. J. Quill, G. LeCroy, Y. Tuchman, H. v. Loo, S. T. Keene, A. Giovannitti, H. R. Lee, I. P. Maria, I. McCulloch, and A. Salleo. Temperature-resilient solid-state organic artificial synapses for neuromorphic computing. Science Advances, 6:eabb2958, 2020. doi: 10.1126/sciadv.abb2958. URL https://doi.org/10.1126/sciadv.abb2958.

[26] Ziyi Guo, Junyao Zhang, Ben Yang, Li Li, Xu Liu, Yutong Xu, Yue Wu, Pu Guo, Tongrui Sun, Shilei Dai, Haixia Liang, Jun Wang, Yidong Zou, Lize Xiong, and Jia Huang. Organic high-temperature synaptic phototransistors for energy-efficient neuromorphic computing. Advanced Materials, 36:2310155, 2024. doi: 10.1002/adma.202310155. URL https://doi.org/10.1002/adma.202310155.

[27] Xueying Wu, Edward Hanson, Nansu Wang, Qilin Zheng, Xiaoxuan Yang, Huanrui Yang, Shiyu Li, Feng Cheng, Partha Pratim Pande, and Janardhan Rao Doppa. Block-wise mixed-precision quantization: Enabling high efficiency for practical ReRAM-based DNN accelerators. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2024. doi: 10.1109/TCAD.2024.3409193. URL https://doi.org/10.1109/TCAD.2024.3409193.

[28] Chenyu Wang, Zhen Dong, Daquan Zhou, Zhenhua Zhu, Yu Wang, Jiashi Feng, and Kurt Keutzer. EPIM: Efficient processing-in-memory accelerators based on epitome. In Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC), pages 1–6, 2024. doi: 10.1145/3649329.3657377. URL https://arxiv.org/abs/2311.07620.

[29] Mengke Ge, Junpeng Wang, Binhan Chen, Yingjian Zhong, Haitao Du, Song Chen, and Yi Kang. Allspark: Workload orchestration for visual transformers on processing in-memory systems. IEEE Transactions on Computers, 2025. doi: 10.1109/TC.2024.3483633. URL https://doi.org/10.1109/TC.2024.3483633.

[30] Cong Wang, Zexin Fu, Jiayi Huang, and Shanshi Huang. Hemlet: A heterogeneous compute-in-memory chiplet architecture for vision transformers with group-level parallelism, 2025. URL https://arxiv.org/abs/2511.15397.

[31] Shiwei Liu, Chen Mu, Hao Jiang, Yunzhengmao Wang, Jinshan Zhang, Feng Lin, Keji Zhou, Qi Liu, and Chixiao Chen. HARDSEA: Hybrid analog-ReRAM clustering and digital-SRAM in-memory computing accelerator for dynamic sparse self-attention in transformer. IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 32(2):269–282, 2024. doi: 10.1109/TVLSI.2023.3337777. URL https://doi.org/10.1109/TVLSI.2023.3337777.

[32] W. Wan, R. Kubendran, B. Gao, S. Joshi, P. Raina, H. Wu, G. Cauwenberghs, and H.-S. P. Wong. A voltage-mode sensing scheme with differential-row weight mapping for energy-efficient RRAM-based in-memory computing. In 2020 IEEE Symposium on VLSI Technology, pages 1–2, 2020. doi: 10.1109/VLSITechnology18217.2020.9265066. URL https://doi.org/10.1109/VLSITechnology18217.2020.9265066.

[33] Xianggao Wang, Na Wei, Shifan Gao, Wenhao Wu, and Yi Zhao. Accuracy improvement with weight mapping strategy and output transformation for STT-MRAM-based computing-in-memory. IEEE Journal on Exploratory Solid-State Computational Devices and Circuits, 10:75–81, 2024. doi: 10.1109/JXCDC.2024.3478360. URL https://doi.org/10.1109/JXCDC.2024.3478360.

[34] Tomas Vincze, Michal Hanic, Martin Berki, and Martin Weis. Simultaneous dual-plasticity organic synaptic transistor for neuromorphic computing. Advanced Electronic Materials, 12(1):e00515, 2026. doi: 10.1002/aelm.202500515. URL https://doi.org/10.1002/aelm.202500515. Published online 2025-12-03.

Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace dataset interfaces. The source data underlying the main-text and Supplementary figures and tables are packaged in the submission-matched source-data archive together with the corresponding manifests, logs, and figure-generation inputs. A reproducibility archive containing the code snapshot, source data, SHA-256 manifest, and `CITATION.cff` file has been staged for Zenodo deposition and will be assigned a DOI at acceptance.

Code Availability

The simulation framework, training and evaluation scripts, device-profile utilities, and plotting code will be provided to editors and reviewers through a reviewer-accessible archive at submission. A curated public release will be made available at https://github.com/Leslie360/HAT upon acceptance. The current repository is distributed under the MIT licence, and the staged Zenodo archive will be updated with the final public DOI at acceptance.

Competing Interests

The authors declare no competing interests.